

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	S. MAGESH	Reg. No.:	192828002
Section / Batch:	D.Tech AIML	Date:	09/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I MAGESH.S (192828002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

36. Application : Search engine indexing

Algorithm used: Divide and Conquer

Algorithm : A search engine contains billions of web pages. Instead of indexing everything at once, the pages are divided into two equal groups. Each group is indexed and merged. It reduces complexity and allows parallel processing.

Recurrence relation $T(n) = 2T(n/2) + n^2$

Identify parameters $a=2$
 $b \neq 2$

(time to combine / process) $F(n) = n^2$

$$n \log_b a = n \log_2^2 = n \Rightarrow F(n) = n^2 > n$$

$$F(n) = \Omega(n^{1+\epsilon})$$

Master theorem case

$$\text{Time Complexity } T(n) = O(n^2).$$

2) Application: AI Based image processing

Algorithm used: Recursive array processing

Algorithm: Large arrays are broken into smaller parts
each part is process recursively and finally
result are combined.

Recurrence relation $T(n) = T(n/2) + \log n$

Parameters $a=1, b=2, F(n) = \log n$

Calculate $n \log_2 1 = 1$

$$\log n = \Omega(1^{+\epsilon})$$

Master Theorem

Time Complexity $= O(\log n)$

3) Application: Image processing system

Algorithm used: Quadtrees / D/C image processing

Algorithm: Large images are recurrence divided into four equal part. each quadrant is processed independently and later combined

Recurrence: $T(n) = 4T(n/2) + n$

Parameters: $a=4$, $b=2$, $F(n)=n$

Calculate: $n \log_2 4 = n^2$, $n = O(n^{2-1})$

Master theorem:

Time Complexity = $O(n^2)$.

4. Application : File Sorting in Distributed System

Algorithm used : Merge Sort

Algorithm : Merge Sort divides the files into two equal levels, sort each half independently and merge the sorted files. It is stable efficient and widely used in distributed computing

Recurrence : $T(n) = 2T(n/2) + n \log n$

Parameters : $a=2, b=2, f(n) = n \log n$

Calculate : $n \log_2^2 = n$

$$f(n) = O(n \log n) = O(n \log_b^a \log n)$$

Master Theorem.

Time Complexity $O(n \log^2 n)$.

5. Application : Genome sequencing analysis

Algorithm used : divide & conquer sequence algorithm

Algorithm : Genome sequencing divides DNA sequences into smaller segments, each segment is analyzed and results are combined to identify mutation and genetic pattern

Recurrence : $T(n) = 3T(n/2) + n$

Parameters : $a=3, b=2, f(n)=n$

Calculate : $n \log_2 3 \approx n^{1.585}$

Since $n = O(n^{1.585 - \epsilon})$

Master theorem

Time Complexity $O(n^{1.585})$